# Choosing Colors

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## Why test a palette

A palette that looks distinct to you can merge for readers with a color vision deficiency. Instead of judging by eye, simulate the common kinds of color vision deficiency and measure how far apart the closest two colors get.

The `colorblindcheck` package (Nowosad 2025) does this. Its `palette_check()` function returns the smallest distance between two colors of a palette, under normal vision and under each simulated deficiency. As a rule of thumb, a smallest distance of about 10 or more means every pair is easy to tell apart. This chapter uses that rule of thumb as its cutoff; the package does not define one. Below the cutoff, some pair is hard to tell apart for some readers.

**Example 0.1** (Comparing qualitative palettes)**.** This example computes, for each palette and number of groups, the smallest distance between two colors under the worst of the three simulated deficiencies (deuteranopia, protanopia and tritanopia). A dash marks a palette that has fewer colors than groups.

```r
options(knitr.kable.NA = "-")

# order puts blue and vermillion first and leaves out black and grey
okabe_ito_order <- c(5, 6, 3, 7, 2, 1, 4)
brewer <- function(name) {
  function(n) RColorBrewer::brewer.pal(max(n, 3), name)[seq_len(n)]
}
palettes <- list(
  "Okabe-Ito (ggokabeito)" = function(n) {
    ggokabeito::palette_okabe_ito(order = okabe_ito_order)[seq_len(n)]
  },
  "RColorBrewer Dark2" = brewer("Dark2"),
  "RColorBrewer Set2" = brewer("Set2"),
  "RColorBrewer Paired" = brewer("Paired"),
  "khroma muted" = function(n) unname(khroma::color("muted")(n)),
  "viridis" = function(n) viridisLite::viridis(n),
  "rcartocolor Safe" = function(n) rcartocolor::carto_pal(n, "Safe")
)

worst_case_distance <- function(make_palette, n) {
  colors <- tryCatch(
    make_palette(n),
    error = function(e) NULL,
    warning = function(w) NULL
  )
  if (is.null(colors) || length(colors) < n || anyNA(colors)) {
    return(NA_real_)
  }
  # viridis adds an alpha channel; palette_check() wants plain "#rrggbb"
  colors <- substr(as.character(colors), 1, 7)
  # the first row is normal vision; the rest are the simulated deficiencies
  min(colorblindcheck::palette_check(colors)$min_dist[-1])
}

expand.grid(palette = names(palettes), groups = c(3, 5, 7, 10)) |>
  tibble::as_tibble() |>
  dplyr::rowwise() |>
  dplyr::mutate(
    distance = worst_case_distance(palettes[[as.character(palette)]], groups)
  ) |>
  dplyr::ungroup() |>
  tidyr::pivot_wider(
    names_from = groups,
    values_from = distance,
    names_prefix = "n = "
  ) |>
  knitr::kable(digits = 1)
```

Table 1: Smallest distance between two colors under simulated color vision deficiency, by palette and number of groups. Higher is better; about 10 or more is easy to tell apart.

| palette | n = 3 | n = 5 | n = 7 | n = 10 |
|---|---|---|---|---|
| Okabe-Ito (ggokabeito) | 12.1 | 12.1 | 11.1 | - |
| RColorBrewer Dark2 | 18.8 | 3.4 | 2.3 | - |
| RColorBrewer Set2 | 12.9 | 1.6 | 1.6 | - |
| RColorBrewer Paired | 8.2 | 8.2 | 4.7 | 1.3 |
| khroma muted | 22.6 | 17.7 | 11.8 | - |
| viridis | 31.6 | 13.7 | 6.8 | 3.9 |
| rcartocolor Safe | 5.8 | 5.8 | 5.8 | 5.3 |

Example 0.1 shows the results. The Okabe-Ito palette (Okabe and Ito 2008) stays above 10 for all 7 of its colors, and with 10 groups no palette tested reaches 10. The RColorBrewer qualitative palettes Dark2 and Set2 are fine for 3 groups and then lose distinctness quickly. The viridis palette scores well for 3 and 5 groups, but it is a sequential palette, so its neighboring colors look alike when the groups have no order. The `khroma` package's Okabe-Ito scheme starts with black, so it is not comparable here. Its `muted` scheme has at most 9 colors.

### What to do with many groups

With 10 groups, no tested palette keeps every pair easy to tell apart, and the table does not test 8 or 9 groups. Pair color with shape or line type, so that color is never the only way to tell groups apart. Better still, show fewer groups: combine small groups, or split the figure into panels.

The `rcartocolor` package's `Safe` palette gives up to 12 colors, so it is the fallback when a figure needs more than 7 groups. Its smallest distance is about 5 to 6 at every size in the table, which is below the cutoff, so it is a fallback and not a recommendation. The `fxtas` package uses it for its biomarker groups.

## References


Nowosad, Jakub. 2025. *Colorblindcheck: Check Color Palettes for Problems with Color Vision Deficiency*. https://CRAN.R-project.org/package=colorblindcheck.

Okabe, Masataka, and Kei Ito. 2008. *Color Universal Design (CUD): How to Make Figures and Presentations That Are Friendly to Colorblind People*. https://jfly.uni-koeln.de/color/.